

An Introduction to Longitudinal Meta-Analysis in R with the metafor Package

Lifespan Social-Personality Preconference
Society for Personality and Social Psychology Annual Convention
Portland, OR

Wolfgang Viechtbauer
Maastricht University
2019-02-07

1

Meta-Analysis: The Basic Idea

- have multiple estimates of some common phenomenon
 - treatment effect
 - association between two variables
 - change in some variable over time
 - the mean of some variable
 - ...
- estimates are usually not equally precise
 - larger sample size → lower variance → higher precision
 - want to give more weight to more precise estimates

2

Standardized Mean Difference

- have means, SDs, and sample sizes for two (independent) groups
- want to quantify the difference between groups

$$d = \frac{\bar{x}_2 - \bar{x}_1}{s_p}$$

where $s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}}$ (pooled SD) or we standardize based on s_1 (or s_2) alone

- sampling variance

$$\text{Var}[d] \approx \frac{1}{n_1} + \frac{1}{n_2} + \frac{d^2}{2(n_1 + n_2)}$$

3

Standardized Mean Change (raw score standardization)

- have means, SDs, and n for a single group at two time points
- want to quantify the change between time points

$$d_r = \frac{\bar{x}_2 - \bar{x}_1}{s_1}$$

- **sampling variance**

$$\text{Var}[d_r] \approx \frac{2(1-r)}{n} + \frac{d_r^2}{2n}$$

- note: need to know the correlation r to compute variance

4

Standardized Mean Change (change score standardization)

- have means, SDs, and n for a single group at two time points
- want to quantify the change between time points

$$d_c = \frac{\bar{x}_2 - \bar{x}_1}{s_c}$$

where $s_c = \sqrt{s_1^2 + s_2^2 - 2rs_1s_2}$ (SD of the change scores)

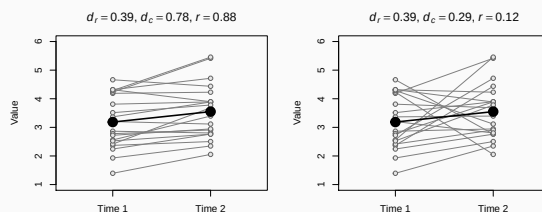
- sampling variance

$$\text{Var}[d_c] \approx \frac{1}{n} + \frac{d_c^2}{2n}$$

5

Difference Between Raw and Change Score Standardization

- **raw score standardization:**
 - **not influenced by rank-order consistency**
 - comparable to standardized mean difference (in principle)
- change score standardization: as $r \rightarrow 1$, d_c increases
 - if $s_1 = s_2$, then $s_c = s_1 \sqrt{2(1-r)}$
 - so if $r = 0.5$, then $d_r = d_c$



6

Meta-Analysis of Longitudinal Studies

- simplest case: each study provides a standardized mean change for a sample measured at age t_1 and at age t_2
- let's create a toy dataset with 6 studies

```
> dat <- data.frame(
+   study = c("Jones et al. (1998)", "Lewis et al. (2004)",
+             "Grant et al. (2006)", "Berry et al. (2013)",
+             "Nolan et al. (2015)", "Clark et al. (2016)"),
+   age1 = c(20, 20, 20, 20, 20, 20),
+   age2 = c(40, 40, 40, 40, 40, 40),
+   mean1 = c(13.4, 2.9, 55.8, 19.2, 6.6, 10.1),
+   mean2 = c(15.1, 3.6, 61.2, 18.8, 8.5, 10.2),
+   sd1 = c(4.8, 1.2, 22.3, 2.9, 3.4, 3.8),
+   n = c(78, 22, 188, 35, 54, 112),
+   r = c(.32, .29, .28, .41, .35, .19))
```

7

Meta-Analysis of Longitudinal Studies

```
> # install metafor package
> install.packages("metafor")

> # load metafor package
> library(metafor)

> # calculate standardized mean changes (with raw score standardization)
> dat <- escalc(measure="SMCR", m1=mean1, m2=mean2,
+               sd1=sd1, ni=n, ri=r, data=dat, slab=study)

> # inspect data frame
> dat
```

##	study	age1	age2	mean1	mean2	sd1	n	r	yi	vi
## 1	Jones et al. (1998)	20	40	13.4	15.1	4.8	78	0.32	0.3507	0.0182
## 2	Lewis et al. (2004)	20	40	2.9	3.6	1.2	22	0.29	0.5622	0.0717
## 3	Grant et al. (2006)	20	40	55.8	61.2	22.3	188	0.28	0.2412	0.0078
## 4	Berry et al. (2013)	20	40	19.2	18.8	2.9	35	0.41	-0.1349	0.0340
## 5	Nolan et al. (2015)	20	40	6.6	8.5	3.4	54	0.35	0.5509	0.0269
## 6	Clark et al. (2016)	20	40	10.1	10.2	3.8	112	0.19	0.0261	0.0145

8

Meta-Analysis of Longitudinal Studies

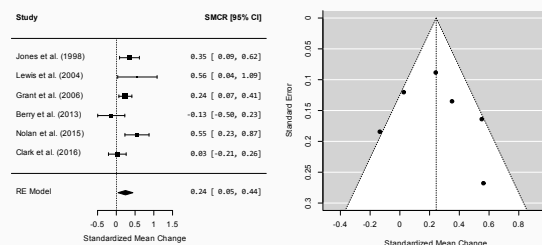
```
> # fit random-effects model and examine results
> res <- rma(yi, vi, data=dat)
> res

## Random-Effects Model (k = 6; tau^2 estimator: REML)
##
## tau^2 (estimated amount of total heterogeneity): 0.0375 (SE = 0.0382)
## tau (square root of estimated tau^2 value): 0.1936
## I^2 (total heterogeneity / total variability): 65.44%
## H^2 (total variability / sampling variability): 2.89
##
## Test for Heterogeneity:
## Q(df = 5) = 12.9747, p-val = 0.0236
##
## Model Results:
##
## estimate se zval pval ci.lb ci.ub
## 0.2444 0.1010 2.4188 0.0156 0.0464 0.4424 *
##
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

9

Meta-Analysis of Longitudinal Studies

```
> # create a forest and funnel plot side-by-side
> # note: the plots below have been customized a bit
> par(mfrow=c(1,2))
> forest(res)
> funnel(res)
```



10

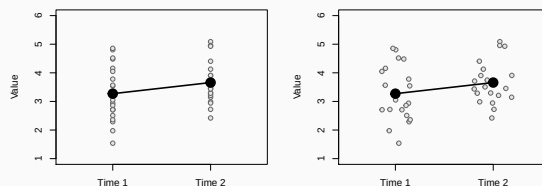
Accounting for Differences in t_1 and/or t_2 Across Studies

- amount of time between the t_1 and t_2 may differ across studies
- one solution:
 - $d_{rs} = d_r / (t_2 - t_1) \times \Delta$
 - $\text{Var}[d_{rs}] = \text{Var}[d_r] / (t_2 - t_1)^2 \times \Delta^2$
 - e.g., $\Delta = 10$ gives change per 10 years
- example:
 - study 1: $d_r = 0.38$ for $t_1 = 20$ and $t_2 = 40$
 $d_{rs} = 0.38 / (40 - 20) \times 10 = 0.19$
 - study 2: $d_r = 0.21$ for $t_1 = 23$ and $t_2 = 35$
 $d_{rs} = 0.21 / (35 - 23) \times 10 = 0.17$
- note: this assumes a constant rate of change within studies

11

Accounting for Differences in t_1 and/or t_2 Across Subjects

- SMC for t_1 to t_2 is something different than SMC for $\bar{t}_1$ to $\bar{t}_2$
- as long as $\text{SD}[t_1]$ and $\text{SD}[t_2]$ are not too large, could ignore this



12

Combining Between- and Within-Subject Designs

- in principle, $d = \frac{\bar{x}_2 - \bar{x}_1}{s_p}$ (between-subject design) is numerically comparable to $d_r = \frac{\bar{x}_2 - \bar{x}_1}{s_1}$ (within-subject design)
- but evidence from within-subject designs is stronger (for measuring change over time) than cross-sectional designs
- analyze separately or code 'design' as a moderator variable and include in the analysis (meta-regression)

13

Unknown Correlation

- a common problem: r is not reported
- some useful equations:

$$r = \frac{s_1^2 + s_2^2 - s_c^2}{2s_1s_2}$$

$$r = 1 - \frac{s_c^2}{2s_1} \quad (\text{if } s_1 = s_2)$$

$$s_c = \frac{\sqrt{n}(\bar{x}_2 - \bar{x}_1)}{t_c}$$

- use a 'guestimate' (based on other studies / own data, reported test-retest correlations, common sense, ...)
- conduct sensitivity analyses using a reasonable range for unknown r values

14

Meta-Analysis of Change over the Lifespan

- each study provides information about two time points

study	t_1	t_2	d_r	var
1	20	24	-.12	.04
2	23	28	.14	.03
3	28	30	.01	.04
4	29	33	.15	.01
5	30	38	.42	.02
6	32	35	.24	.03

15

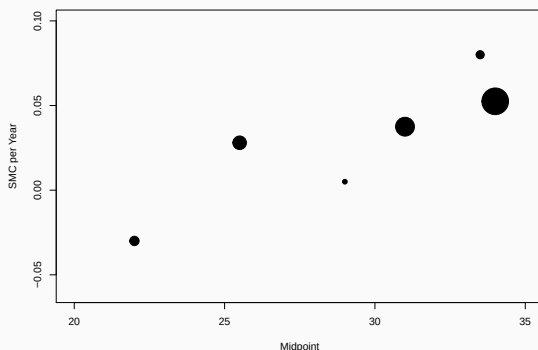
Meta-Analysis of Change over the Lifespan

- let's create this toy dataset

```
> dat <- data.frame(
+   study = c("Jones et al. (1998)", "Lewis et al. (2004)",
+             "Grant et al. (2006)", "Berry et al. (2013)",
+             "Nolan et al. (2015)", "Clark et al. (2016)"),
+   age1 = c( 20, 23, 28, 29, 30, 32),
+   age2 = c( 24, 28, 30, 33, 38, 35),
+   yi = c(-.12, .14, .01, .15, .42, .24),
+   vi = c(.04, .03, .04, .01, .02, .03))
>
> # compute standardized mean change per year
> dat$yi <- with(dat, yi / (age2 - age1))
> dat$vi <- with(dat, vi / (age2 - age1)^2)
>
> # calculate midpoint of each interval
> dat$mage <- with(dat, (age1 + age2) / 2)
```

16

Meta-Analysis of Change over the Lifespan



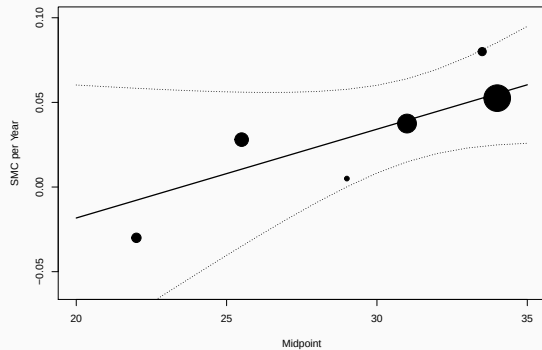
17

Meta-Analysis of Change over the Lifespan

```
> res <- rma(yi, vi, mods = ~ mage, data=dat)
> res
## Mixed-Effects Model (k = 6; tau^2 estimator: REML)
##
## tau^2 (estimated amount of residual heterogeneity): 0 (SE = 0.0007)
## tau (square root of estimated tau^2 value): 0
## I^2 (residual heterogeneity / unaccounted variability): 0.00%
## H^2 (unaccounted variability / sampling variability): 1.00
## R^2 (amount of heterogeneity accounted for): 0.00%
##
## Test for Residual Heterogeneity:
## QE(df = 4) = 0.7626, p-val = 0.9434
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 2.4234, p-val = 0.1195
##
## Model Results:
##
##           estimate      se      zval      pval      ci.lb      ci.ub
## intrcpt    -0.1232   0.1062   -1.1599   0.2461   -0.3314   0.0850
## mage         0.0052   0.0034    1.5567   0.1195   -0.0014   0.0118
```

18

Meta-Analysis of Change over the Lifespan



19

Studies with More Than Two Time Points

- for example:

age	20	24	28
mean	$\bar{x}_1$	$\bar{x}_2$	$\bar{x}_3$
SD	s_1	s_2	s_3

- can compute $d_{r_{12}} = \frac{\bar{x}_2 - \bar{x}_1}{s_1}$ and $d_{r_{23}} = \frac{\bar{x}_3 - \bar{x}_2}{s_2}$
- but $d_{r_{12}}$ and $d_{r_{23}}$ are not independent
- need to account for their covariance
- also: since we now have multiple estimates from the same study, need to use a multilevel meta-analysis model
- the details are beyond the purposes of this talk

20

References

- Becker, B. J. (1988). Synthesizing standardized mean-change measures. *British Journal of Mathematical and Statistical Psychology*, 41(2), 257-278.
- Gibbons, R. D., Hedeker, D. R., & Davis, J. M. (1993). Estimation of effect size from a series of experiments involving paired comparisons. *Journal of Educational Statistics*, 18(3), 271-279.
- Ishak, K. J., Platt, R. W., Joseph, L., Hanley, J. A., & Caro, J. J. (2007). Meta-analysis of longitudinal studies. *Clinical Trials*, 4(5), 525-539.
- Morris, S. B. (2000). Distribution of the standardized mean change effect size for meta-analysis on repeated measures. *British Journal of Mathematical and Statistical Psychology*, 53(1), 17-29.
- Morris, S. B. (2008). Estimating effect sizes from pretest-posttest-control group designs. *Organizational Research Methods*, 11(2), 364-386.
- Morris, S. B., & DeShon, R. P. (2002). Combining effect size estimates in meta-analysis with repeated measures and independent-groups designs. *Psychological Methods*, 7(1), 105-125.
- Musekiwa, A., Manda, S. O., Mwambi, H. G., & Chen, D. G. (2016). Meta-analysis of effect sizes reported at multiple time points using general linear mixed model. *PLOS ONE*, 11(10), e0164898.
- Peters, J. L., & Mengersen, K. L. (2008). Meta-analysis of repeated measures study designs. *Journal of Evaluation in Clinical Practice*, 14(5), 941-950.
- Trikalinos, T. A., & Olkin, I. (2012). Meta-analysis of effect sizes reported at multiple time points: A multivariate approach. *Clinical Trials*, 9(5), 610-620.
- Wolff Smith, L. J., & Beretvas, S. N. (2009). Estimation of the standardized mean difference for repeated measures designs. *Journal of Modern Applied Statistical Methods*, 8(2), 600-609.

21

wolfgang.viechtbauer@maastrichtuniversity.nl

wvb@wvbauer.com

<http://www.wvbauer.com/>

<http://www.metafor-project.org/>

[@wviechtb](#)

22